



Sterilizing drinking water with sunlight

Researchers in China have come up with a method for eliminating bacteria in drinking water using a special material and a bit of sunlight. <https://digit.in/mar19-21>



Grenade with potatoes

WWI German grenade was found in a Hong Kong chip factory among potatoes imported from France. <https://digit.in/mar19-22>

magnitude as compared to the typical arrangement. What that means is that the task described above can be performed by such an array in a better way and also in the presence of a lot of noise in the signal.

METHOD

To pull this off, the design uses something known as 'heterodyne detectors'. To understand its significance, you have to understand what heterodyning is. It is a signal processing technique that involves using two signals to create a third one, usually by combining or mixing. This results in the shifting of one frequency range into another. To put it simply, say there are two signals f_1 and f_2 that are mixed, the resulting new signals will be created at f_1+f_2 and f_1-f_2 . Typically one of these is enough for a particular purpose, and the other is discarded via a filter.

For this particular design, heterodyne detectors were fitted in a compact array, for which the researchers had to drastically reduce the size of said detectors. It would fulfill three purposes at the same time – down-mix input signals, synchronize the pixel array, and produce strong output baseband signals.

DEVICE

A prototype was built with a 32-pixel array integrated on a 1.2-square-millimeter device. Compared to the currently state of the art sub-terahertz on-chip array sensors, the pixels on this prototype were 4300 times more sensitive. "A big motivation for this work is having better 'electric eyes' for autonomous vehicles and drones," says co-author Ruonan Han, an associate professor of electrical engineering and computer science, and director of the Terahertz Integrated Electronics Group in the MIT Microsystems Technology Laboratories (MTL). "Our low-cost, on-chip sub-terahertz sensors will play a complementary role to LiDAR for when the environment is rough." According to him, with further development this device would be perfect for autonomous vehicles and robots.

Essentially, for a particular signal, the 'heterodyne pixel' generates a frequency beat, which is the frequency difference between two incoming sub-terahertz signals (f_1-f_2). Additionally, an electrical component known as the local oscillation is generated, which changes the frequency of an input frequency. A "down-mixing" occurs which produces a signal in the megahertz range that is easily detectable by the baseband processor in the design.

SYNCHRONISATION

The output signal being detected here can tell the distance of an object, similar to what LiDAR does. Additionally, if the outputs of multiple such pixels in an array is aligned, synchronised and detected, it is also possible to generate high-resolution

shared among more pixels, leading to a weaker output signal and subsequently, poor detection.

DECENTRALISATION

The researchers have decided to solve this issue by bringing the signal generation aspect to the pixels themselves. The local oscillation signal, mentioned earlier as a component of the down mixing process, is generated by each individual pixel. Additionally, a coupler is included that is responsible for synchronising the local oscillation signal of a pixel with its neighbour. This also standardises the output power, since a single source is not being divided.

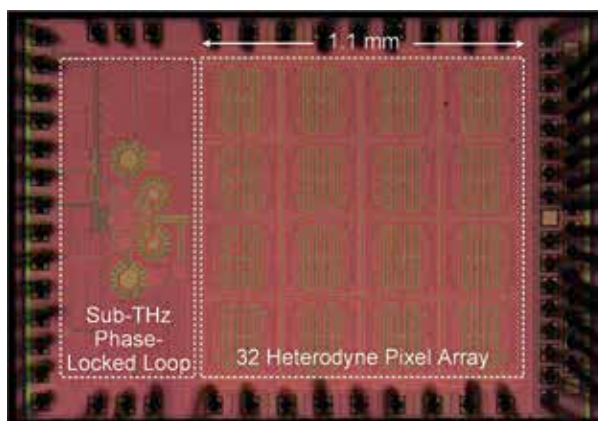
This design, however, poses another challenge. Bringing the capabilities to the pixel itself makes the footprint

of each pixel larger, which makes it difficult for them to be integrated into a compact large-scale high-density array. To overcome this, the researchers have included a multitasking component for each pixel that takes over the responsibilities of four traditionally separate components

- antenna, downmixer, oscillator, and coupler – and reduces the overall footprint, allowing the decentralised design to be executed.

STABILITY

To improve the stability of the overall system, the researchers included a phase-locked loop into their design. This component provides a reference to all the 32 local oscillation signals to lock their sub-terahertz frequency on to. Due to the coupling between the pixels, the signals at each pixel are essentially identical in their high-stability phase and frequency. This entire architecture minimizes signal loss and maximizes control. 



The chip design

images of a scene. This enhances the capability of the system beyond detection and into object recognition, a crucial aspect of vision in autonomous vehicles and robots.

For this to work, the signals from all the pixels in the heterodyne array have to be synchronised, possibly using some signal synchronisation technique. Usually, a single hub is employed that shares a local oscillation signal with all the individual pixels. However, in this particular design it would cause a problem. Generating a high power signal from a single hub is not easy, but a requirement in this approach. Scaling up is an issue, since the power from the single hub will be